

Trait-based Ecology III

Today's Agenda:

- Quiz
- Traits and ecosystem function

Syllabus update

Major things remaining:

- 1) Project 3 (Due Friday)
- 2) Group project (due Wed, April 29th in class)
- 3) Final: Wed, May 6th from 12:30 to 2:30.
- 4) Project 4 (due May 7th, end of semester)

Group Assignment

- Focus: variable/changing ecosystems in Florida
- E.g., variation due to:
 - Ecological succession
 - Temporally fluctuating environments
 - Global changes
- Some obvious examples might be:
 - Fire-dependent ecosystems
 - Tidal systems
 - Hurricanes as successional drivers
 - Ephemeral water bodies

Group Assignment

- Pick your own group (Groups can be of one person)
- Everyone in a group gets the same grade
- Do/make something that explains your your chosen system to lay people
- Can be anything that can convey a message
 - Possibly bonus points for creativity

Group Assignment

- Products will be presented in class on April 29th
- Show off what you created to the class (and me)
- Your group should briefly explain:
 - what you did/created
 - rationale behind the approach you took
 - how it conveys the underlying concepts

Questions so far?

Project 4

Due May 7th before midnight

- University policies force me to be a bit stricter about the deadlines on this one
- Feel free to submit early to avoid the end-of-semester crunch!

Project 4

- Pick a research question about community ecology
- Propose alternative research approaches
- Three parts:
 - 1) Intro (1-2 pages, 4 max)
 - 2) Methods (**funded** version, 1-2 pages)
 - 3) Methods (**free** version, 1-2 pages)

Project 4

- Ecology is cool because it can often be done for free
- BUT you can usually do even better work with some funding
- Goal here is to think about different approaches to a question

Project 4

- No set number of references to read
- Re-use as much old text as you like
- Remember to cite where you need to
- Worth 15 points (other projects were 10)

Questions on Project 4?

Next week:

I'll be out of town, so no class.

- I'll be cutting the lectures on Metacommunities
 - But they're posted if you really want to read them...
- Full points to everyone for next week's quizzes
- I recommend you use the time to work on the group project

Questions?

Why focus on traits?

- Generality:
 - Rather than focusing on particular species in particular systems, we can say things about all species in all systems (or at least a larger subset of species/systems).
- Quantitative:
 - We move from species (categorical) to continuous measurements
- Links to mechanism
 - We can focus on traits that are linked to particular mechanisms
 - E.g., if we want to work on predation, mouth size might be useful.
- More predictive
 - Allow us to say something about potential future scenarios

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Traits and ecosystem function

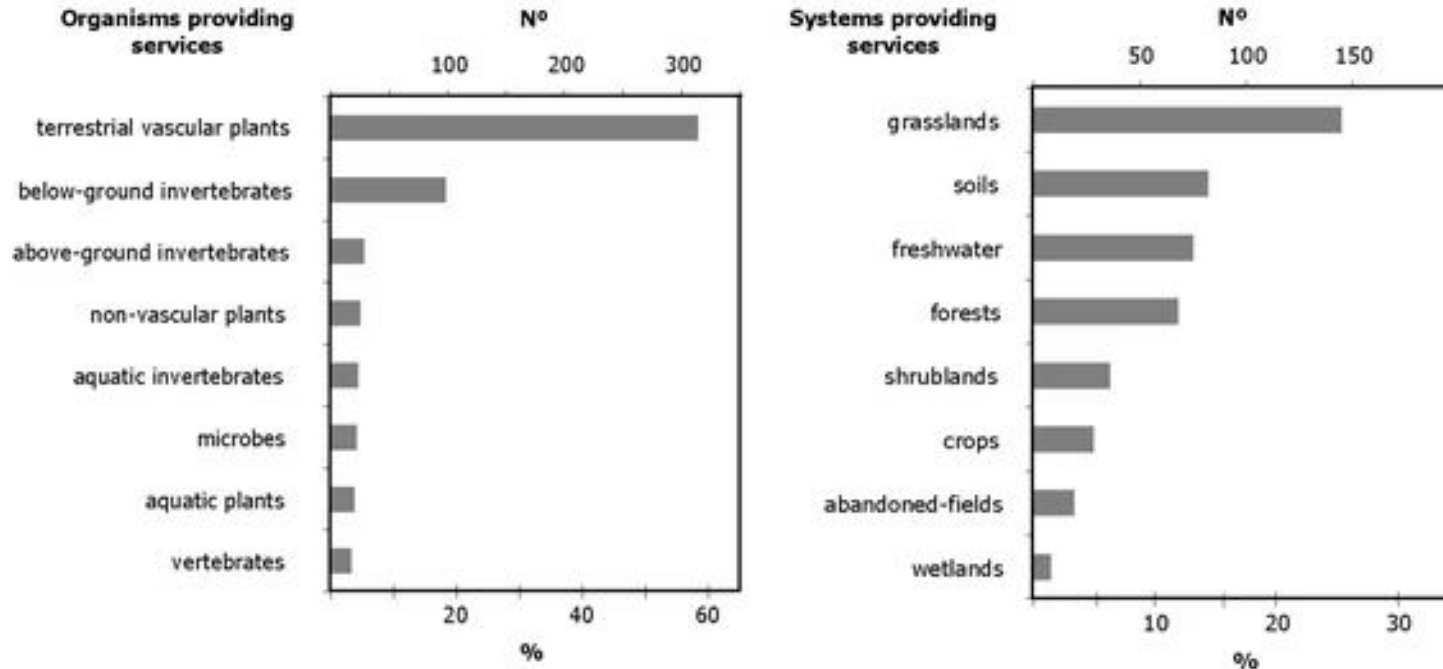
- Which organisms?
- Which traits map onto which services?
- How much do traits vs environment matter?

Towards an assessment of multiple ecosystem processes and services via functional traits

Francesco de Bello · Sandra Lavorel · Sandra Díaz · Richard Harrington · Johannes H. C. Cornelissen · Richard D. Bardgett · Matty P. Berg · Pablo Cipriotti · Christian K. Feld · Daniel Hering · Pedro Martins da Silva · Simon G. Potts · Leonard Sandin · Jose Paulo Sousa · Jonathan Storkey · David A. Wardle · Paula A. Harrison

What are people studying?

RANKING OF ORGANISMS AND SYSTEMS PROVIDING SERVICES AS FOUND IN THE LITERATURE

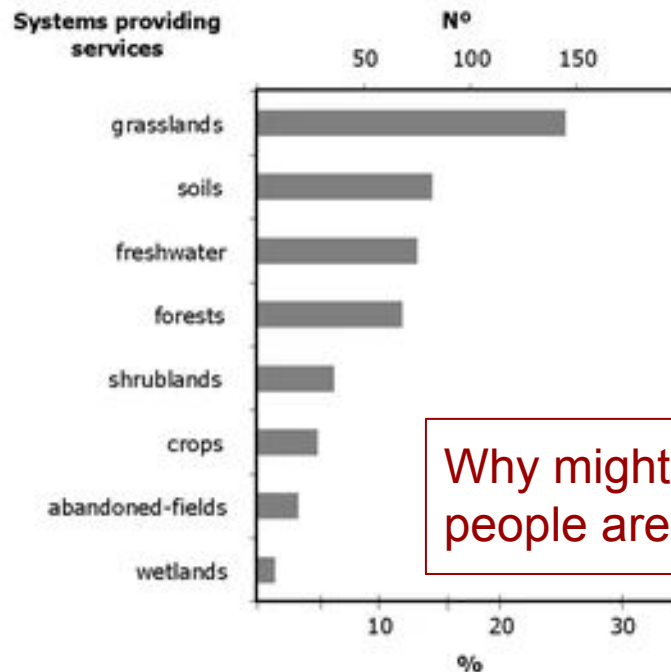
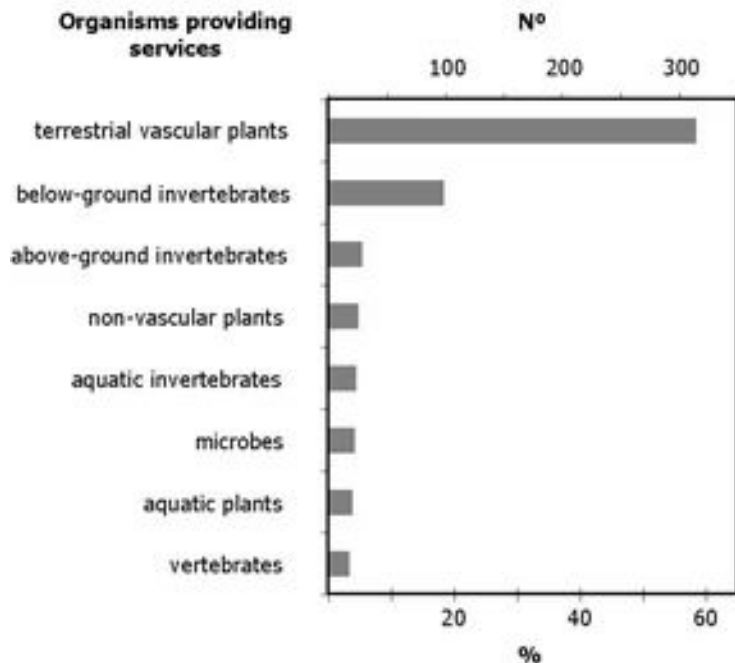


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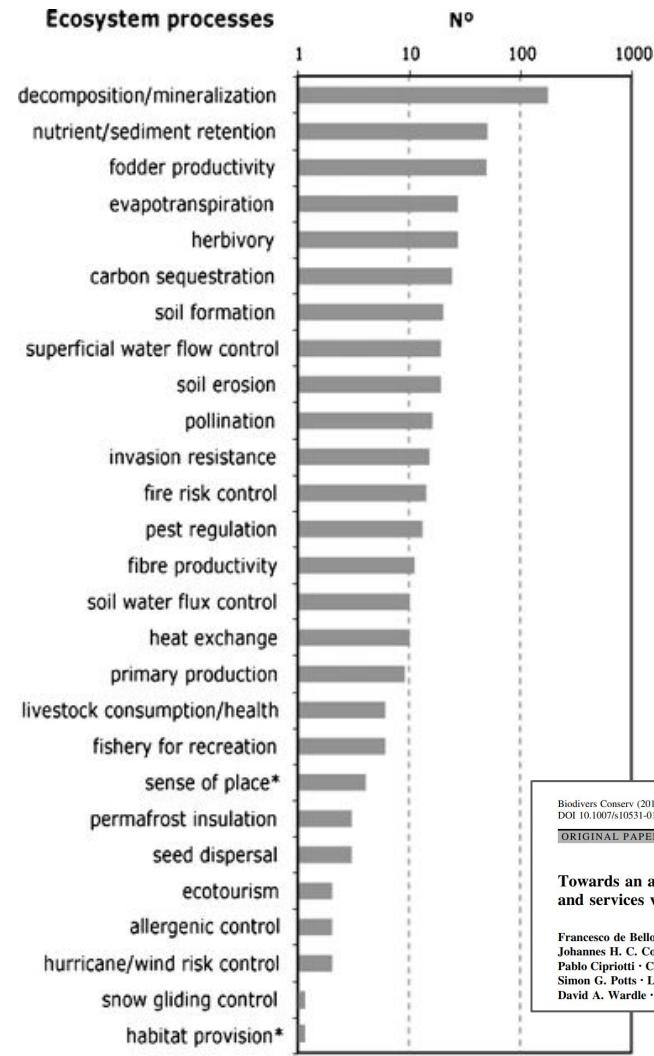
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Why might these be what people are studying?

What are people studying?



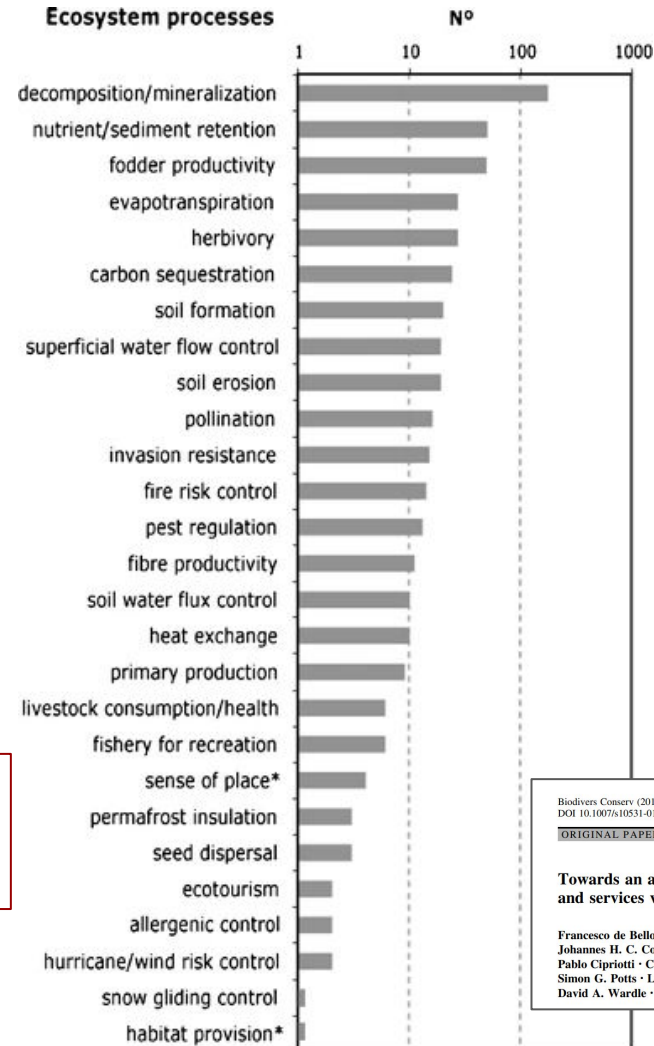
Biodivers. Conserv. (2010) 19:2873–2893
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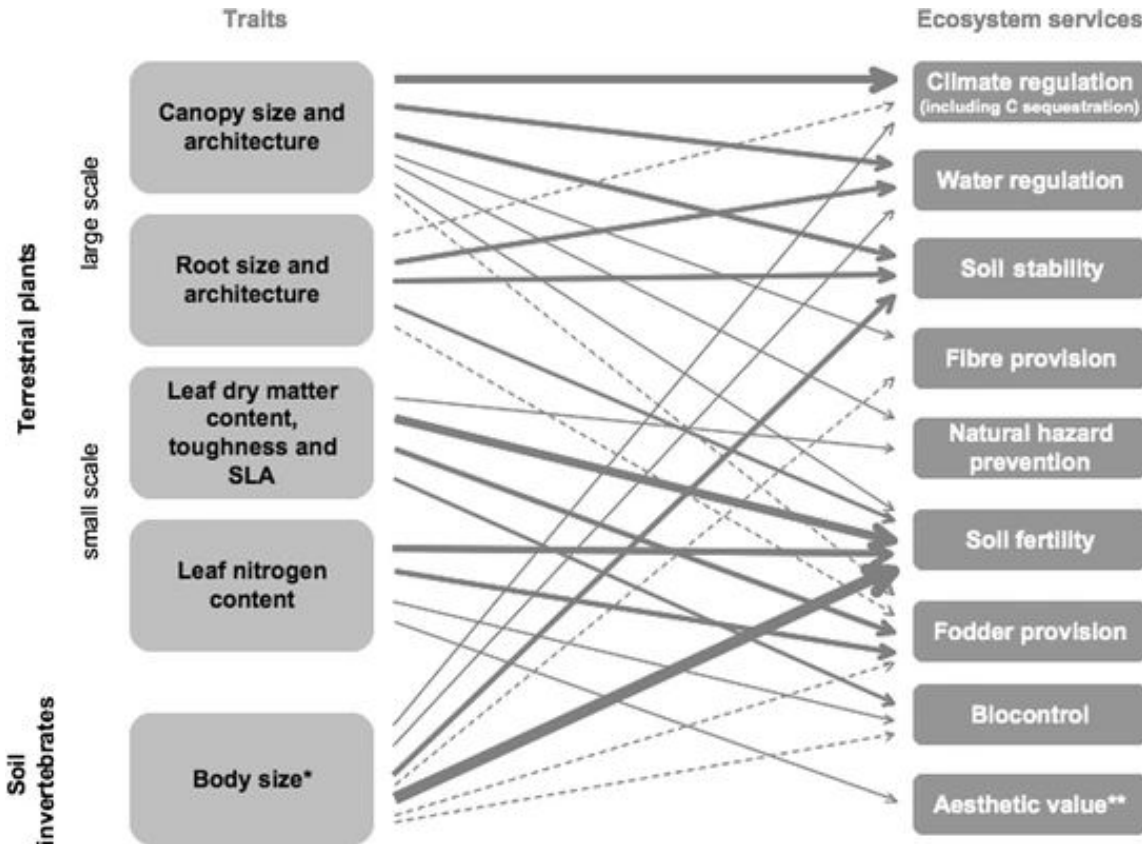
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What are people studying?

Traits

Ecosystem services



Ecosystem function axes

- Lots of ecosystem functions we might study
- Some correlated
- Wanted to understand major variation

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

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 Check for updates

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Ecosystem function axes

Ecosystem services measured:

- WUEt, variant of water-use efficiency, but based on transpiration estimates
- underlying water-use efficiency (uWUE)
- maximum and mean basal ecosystem respiration (Rbmax and Rb, respectively)
- maximum net ecosystem productivity (NEPmax),
- maximum dry canopy surface conductance (Gsmax)
- maximum gross CO2 uptake at light saturation (GPPsat)

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Ecosystem function axes

Continued:

- stomatal slope at ecosystem scale (G1)
- maximum evapotranspiration (ET_{max}),
- EF amplitude (EF_{ampl})
- evaporative fraction (EF) (that is, the ratio between latent heat flux and available energy, indicative of energy partitioning)
- apparent carbon-use efficiency (aCUE) (that is, the remaining fraction of carbon entering the ecosystem).

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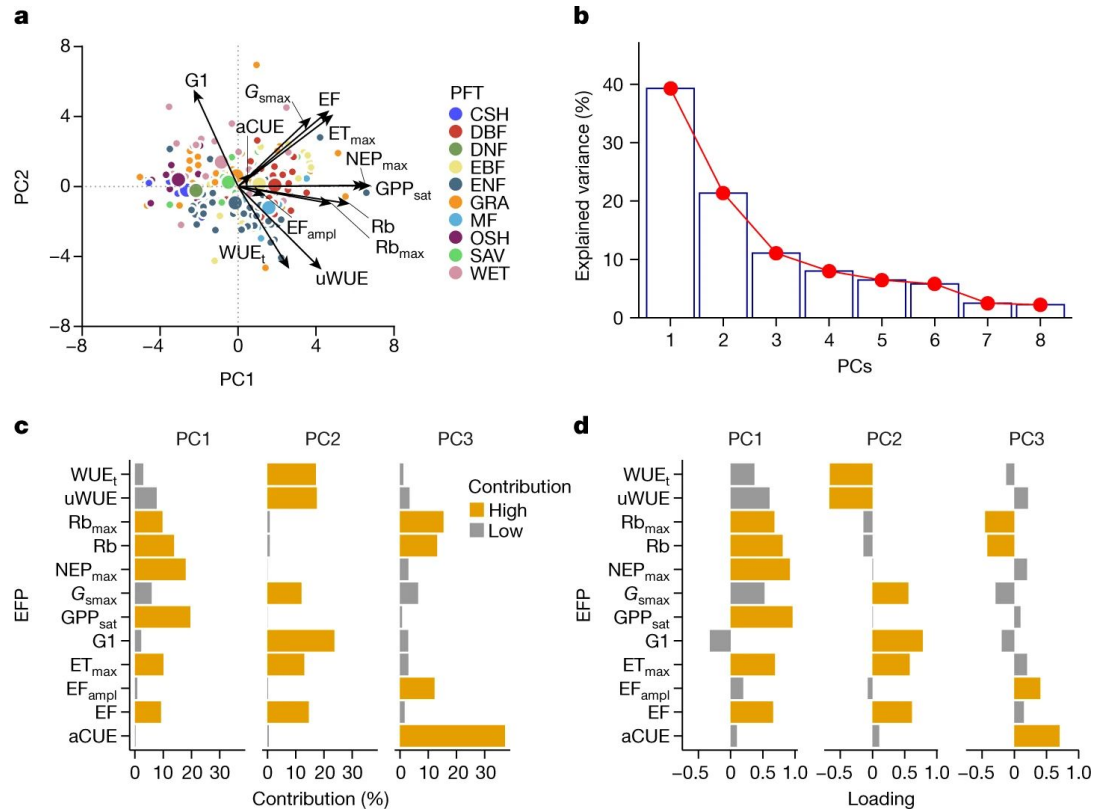
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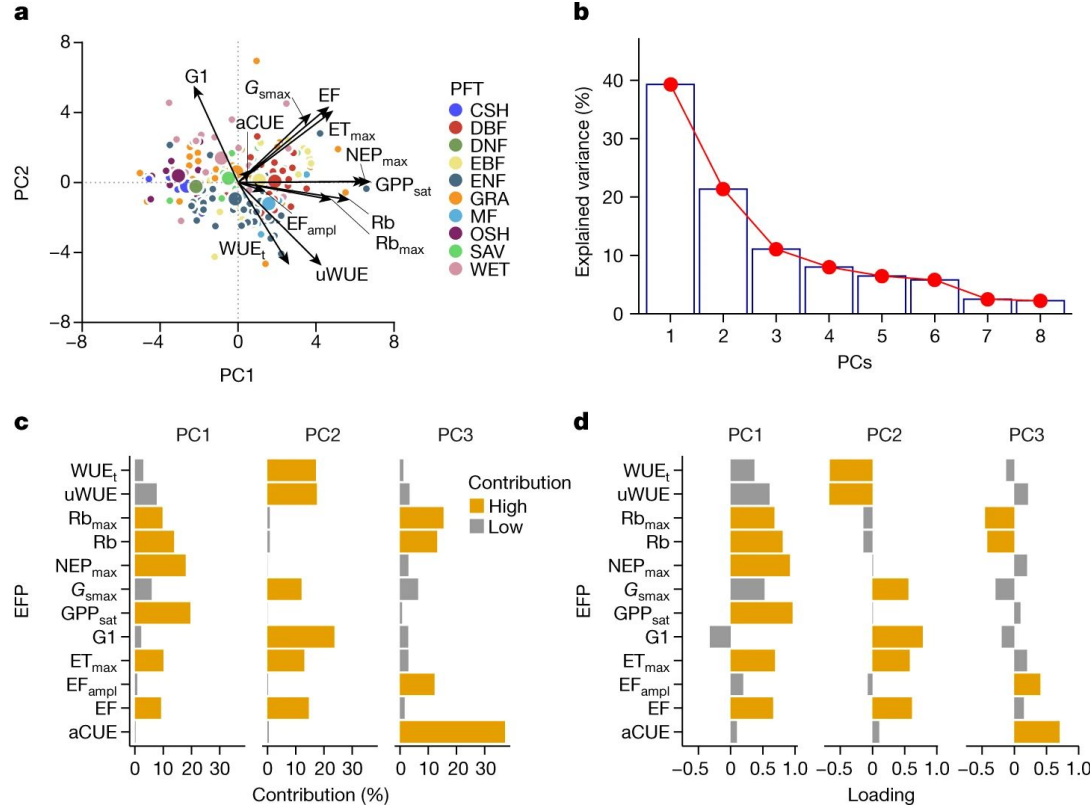
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Ecosystem function axes



PC1 = maximum ecosystem productivity
 PC2 = ecosystem water-use strategies
 PC3 = ecosystem carbon-use efficiency

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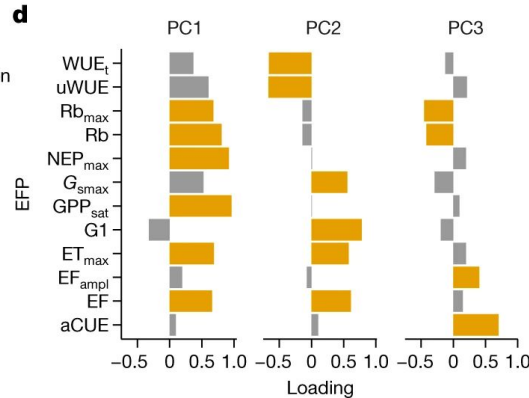
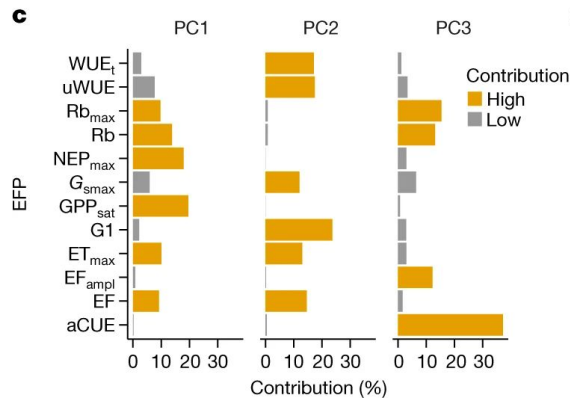
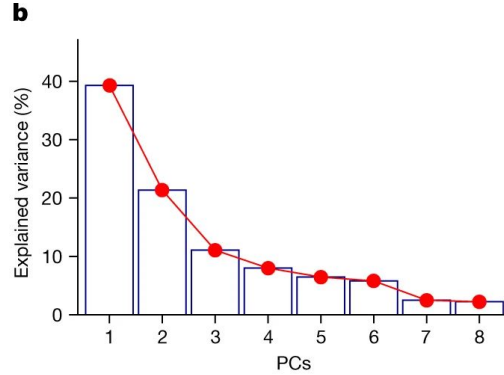
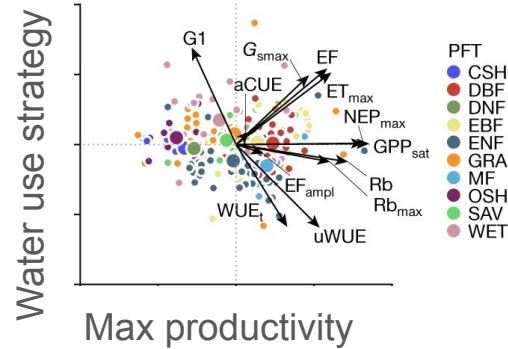
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Ecosystem function axes



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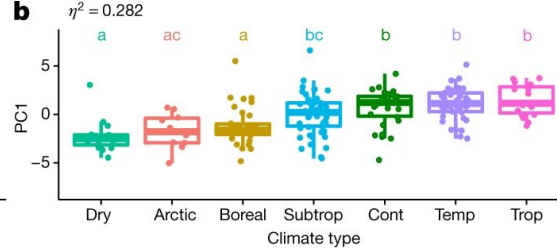
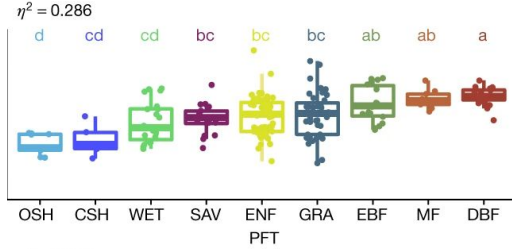
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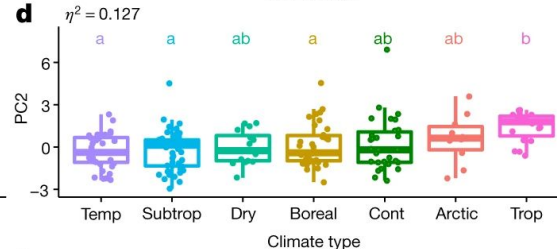
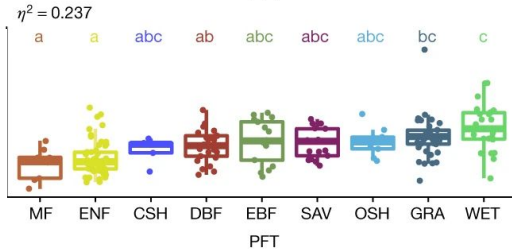
CSH = closed shrublands
 DBF = deciduous broadleaved forest
 DNF = deciduous needleleaf forests
 EBF = evergreen broadleaved forest
 ENF = evergreen needleleaf forest
 GRA = grasslands
 MF = mixed forest
 OSH = open shrublands
 SAV = savannah
 WET = wetlands

Ecosystem function axes

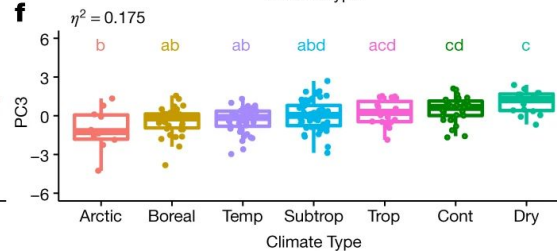
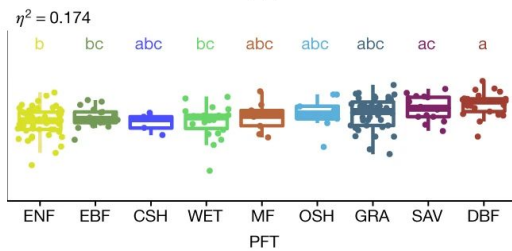
Max productivity



Water use



Carbon use



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DBF = deciduous broadleaved forest
DNF = deciduous needleleaf forests
EBF = evergreen broadleaved forest
ENF = evergreen needleleaf forest
GRA = grasslands
MF = mixed forest
OSH = open shrublands
SAV = savannah
WET = wetlands

Ecosystem function axes

What traits best explain this variation?

- Compared plant traits vs climate traits

Article

The three major axes of terrestrial ecosystem function

<https://doi.org/10.1038/s41586-021-03939-9>

Received: 30 October 2019

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 Check for updates

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Ecosystem function axes

Plant traits:

- Nitrogen content (N%)
- maximum leaf area index (LAI_{max})
- maximum canopy height (H_c)
- above-ground biomass (AGB)

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Ecosystem function axes

Climate traits:

- cumulative soil water availability index (CSWI),
- mean annual precipitation (P)
- mean shortwave incoming radiation (SWin)
- mean air temperature (Tair)
- mean vapour pressure deficit (VPD)

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Ecosystem function axes

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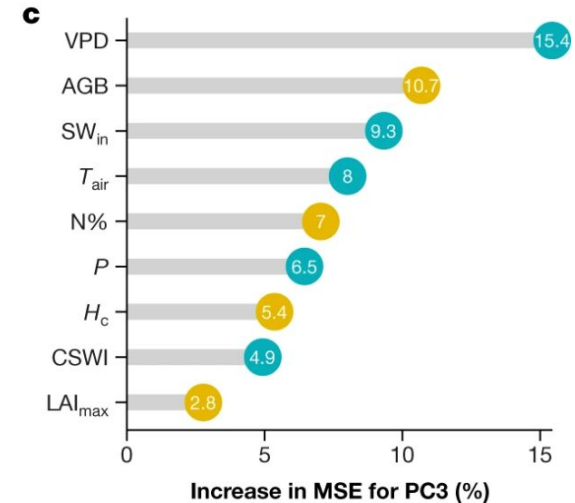
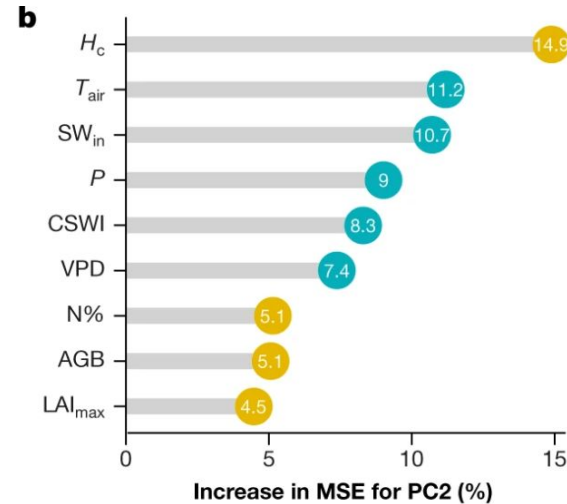
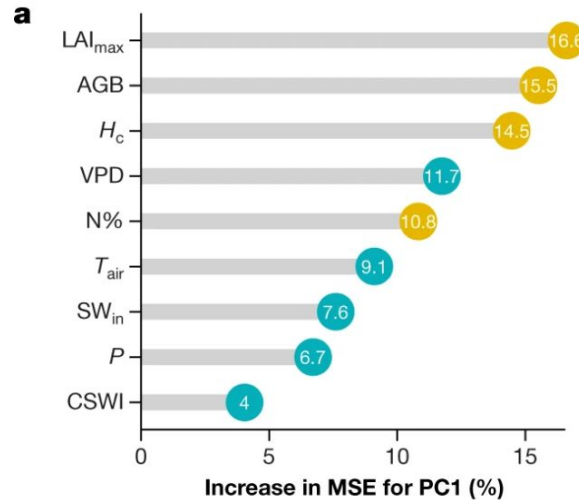
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PC1 = maximum ecosystem productivity PC2 = ecosystem water-use strategies PC3 = ecosystem carbon-use efficiency

Ecosystem function axes

Note:

These types of studies are always sensitive to which traits, which services, etc.

Still, shows how important traits can be for linking with function

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Which organisms are most important?

Mass ratio hypothesis:

- The traits of the species that account for most of the biomass drive functioning
- Grime noted that that this only refers to autotrophs

*Journal of
Ecology* 1998,
86, 902–910

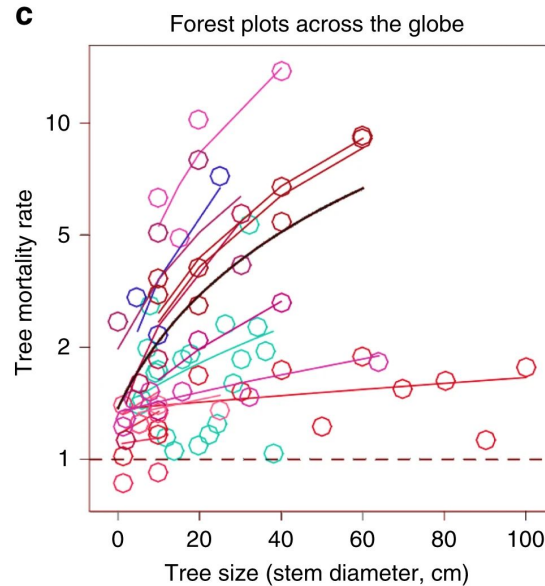
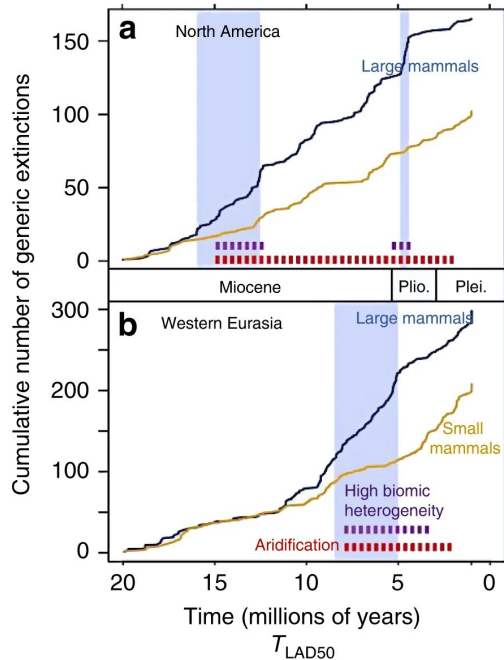
ESSAY REVIEW

Benefits of plant diversity to ecosystems: immediate, filter and founder effects

J.P. GRIME

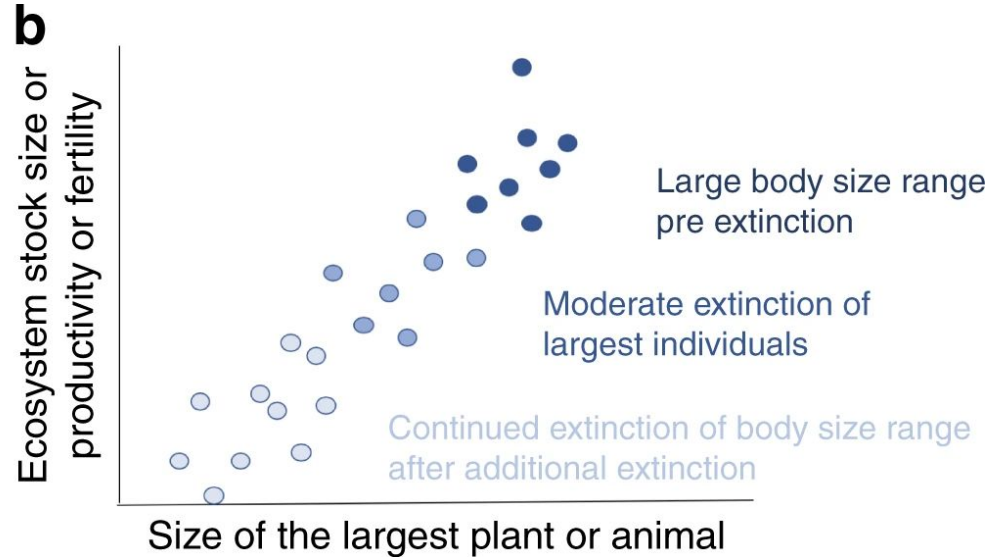
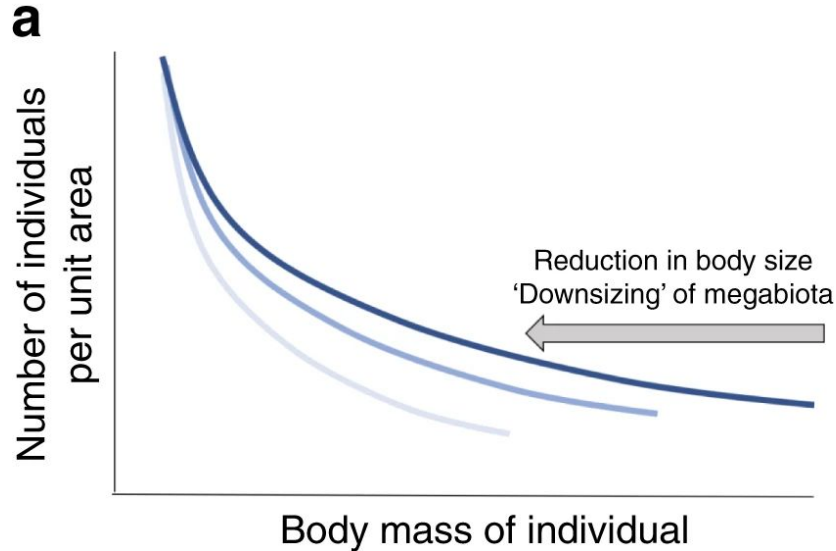
Which organisms are most important?

The big things drive functions



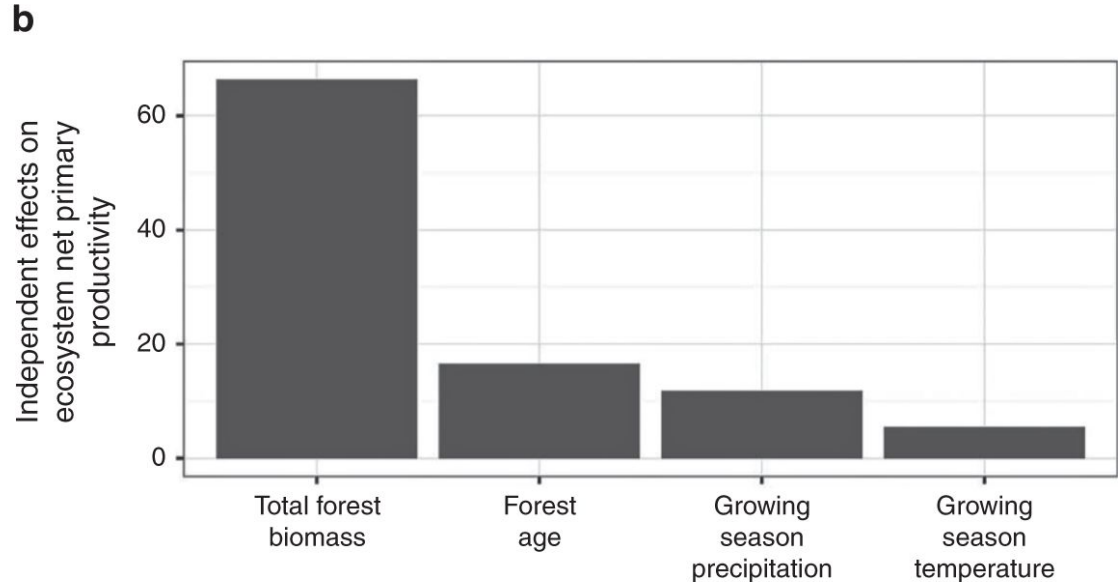
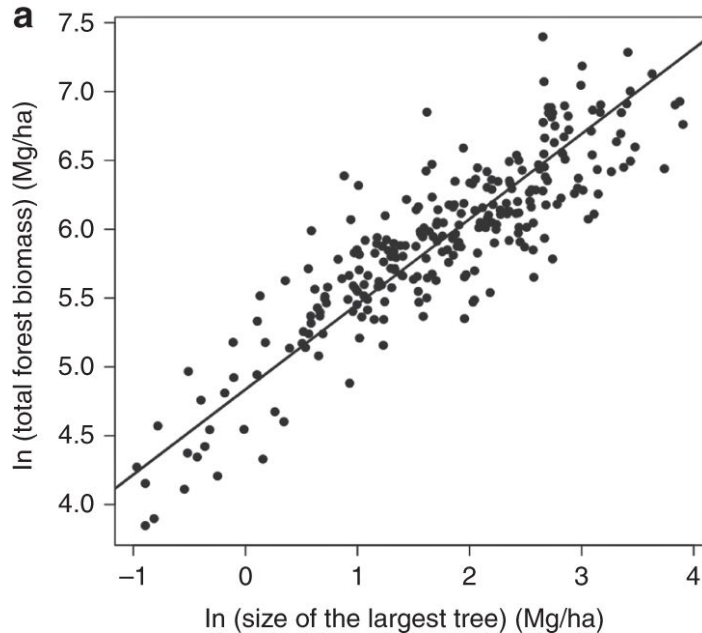
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The big things drive functions



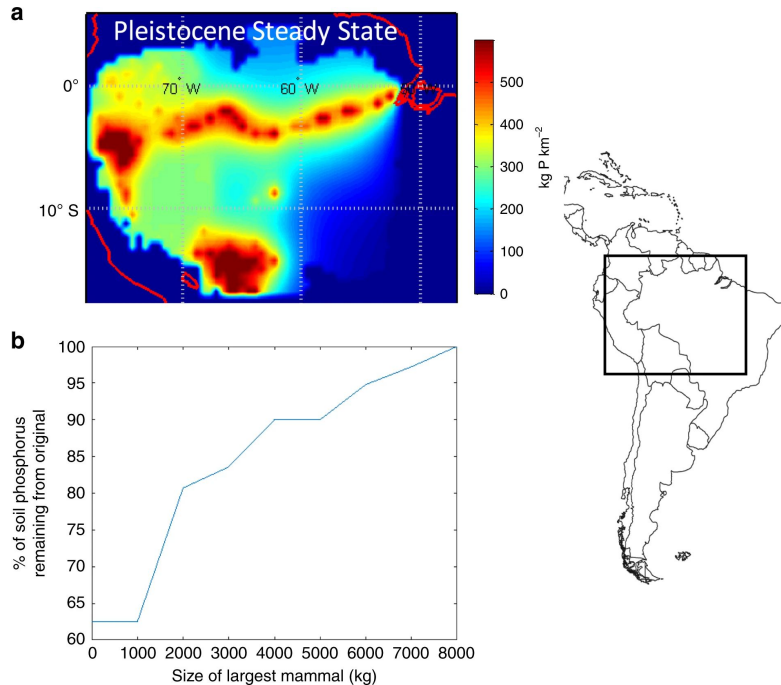
Which organisms are most important?

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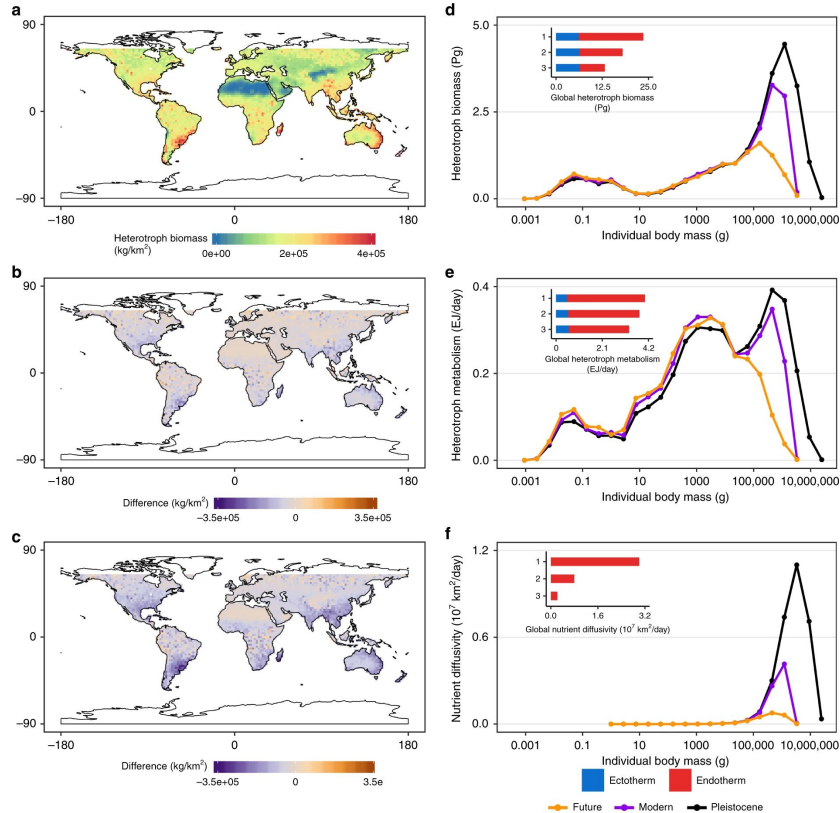


Which organisms are most important?

The big things drive functions



Which organisms are most important?



Megafauna services

Q: How has megafauna loss impacted nutrients?

Megafauna services

Q: How has megafauna loss impacted nutrients?

Approach: models based on traits

Traits used:

- Body size (M)
- Range size (DD)
- Metabolic rate (MR)
- Population density (PD)
- Food passage time (PR)
- Total plant biomass (B)
- Edible fraction of plant biomass (α)

Megafauna services

Q: How has megafauna loss impacted nutrients?

Approach: models based on traits

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$$\Phi = (1 - \epsilon) * MR * \frac{PD}{\alpha B} * \frac{(DD * PR)^2}{2 * PR} = \frac{0.78 * 0.05 * M^{1.17}}{\alpha B} \quad (1)$$

Megafauna services

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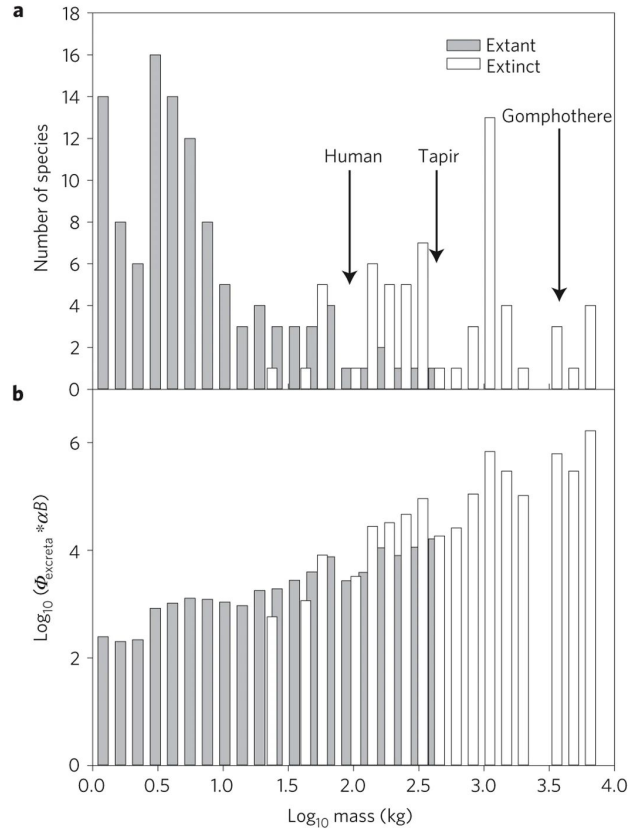
$$\Phi = (1 - \epsilon) * MR * \frac{PD}{\alpha B} * \frac{(DD * PR)^2}{2 * PR} = \frac{0.78 * 0.05 * M^{1.17}}{\alpha B} \quad (1)$$

**Note: superlinear scaling
means bigger things are
disproportionately important!**

Megafauna services

The legacy of the Pleistocene megafauna extinctions on nutrient availability in Amazonia

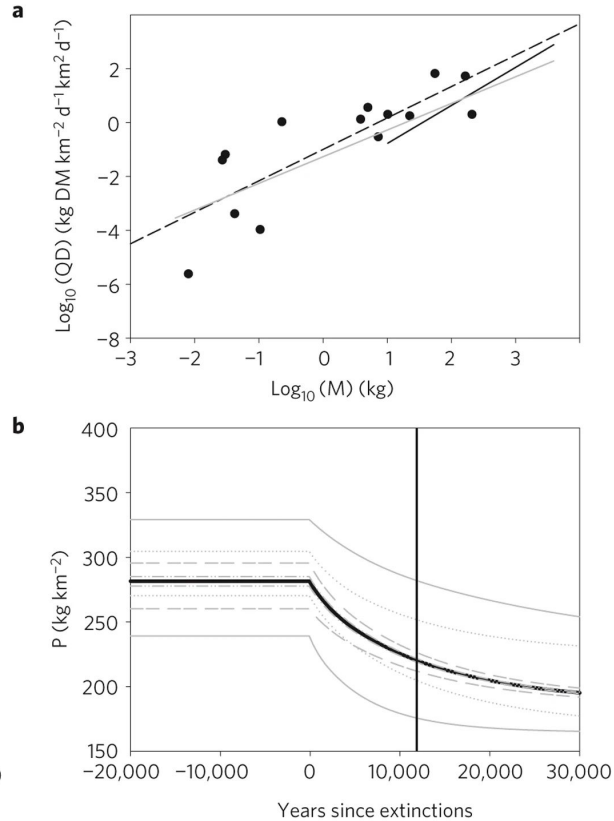
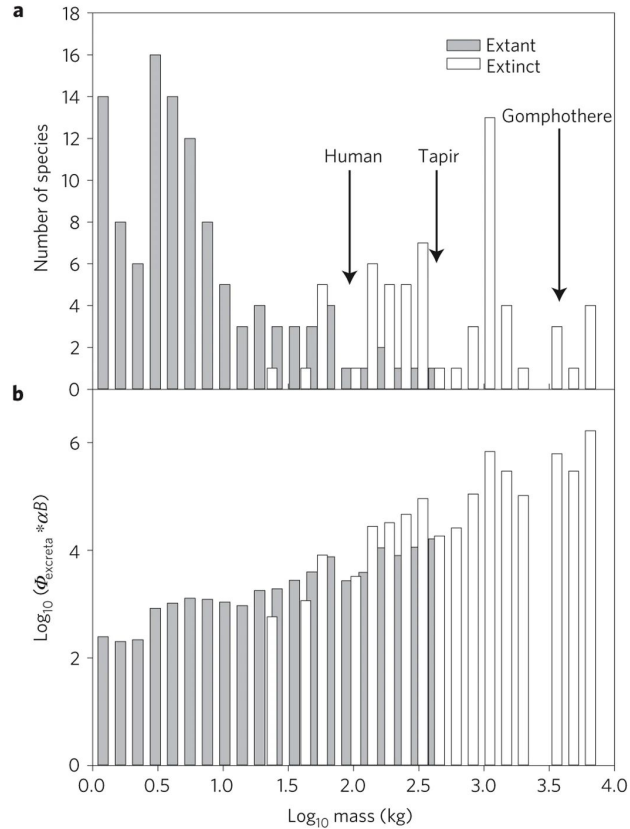
Christopher E. Doughty¹*, Adam Wolf² and Yadvinder Malhi¹



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Christopher E. Doughty^{1*}, Adam Wolf² and Yadvinder Malhi¹

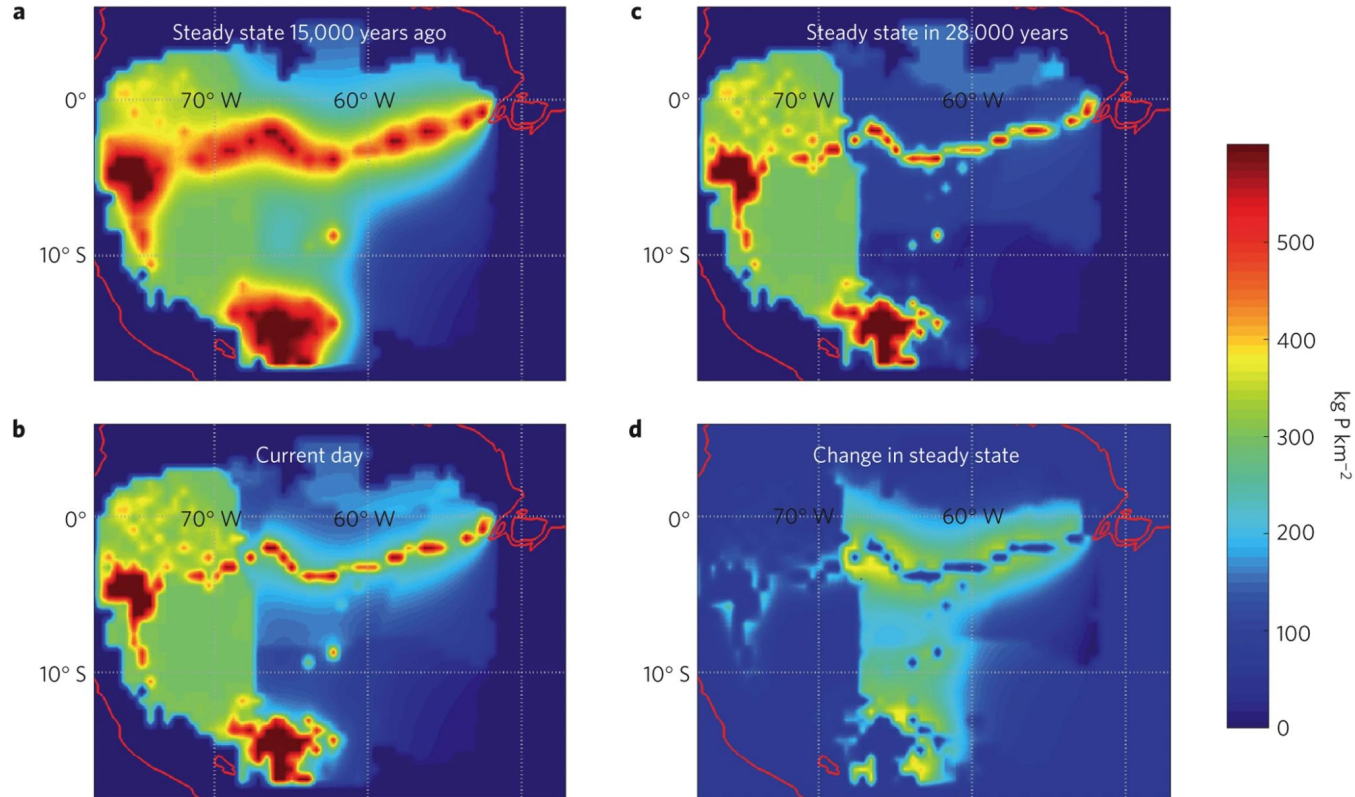
Megafauna services



Megafauna services

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extinctions on nutrient availability in Amazonia

Christopher E. Doughty^{1*}, Adam Wolf² and Yadvinder Malhi¹

Megafauna services

Table 1 | Average $\Phi_{\text{excreta}} * \alpha B$ ($\text{km}^2 \text{yr}^{-1}$) for each continent calculated for modern species and modern plus extinct species.

	North America	South America	Australia	Eurasia	Africa
Number of species extinct	65	64	45	9	13
Mean weight of extinct animals (kg)	846	1,156	188	2,430	970
Modern $\Phi_{\text{excreta}} * \alpha B$	13,876	12,934	21,804	21,779	265,621
Modern + extinct fauna $\Phi_{\text{excreta}} * \alpha B$	140,716 ($\pm 38,000$)	283,854 ($\pm 81,000$)	48,250 ($\pm 8,000$)	118,349 ($\pm 29,000$)	324,848 ($\pm 18,000$)
Percentage of original	10% ($\pm 2\%$)	5% ($\pm 1\%$)	45% ($\pm 6\%$)	18% ($\pm 4\%$)	82% ($\pm 4\%$)

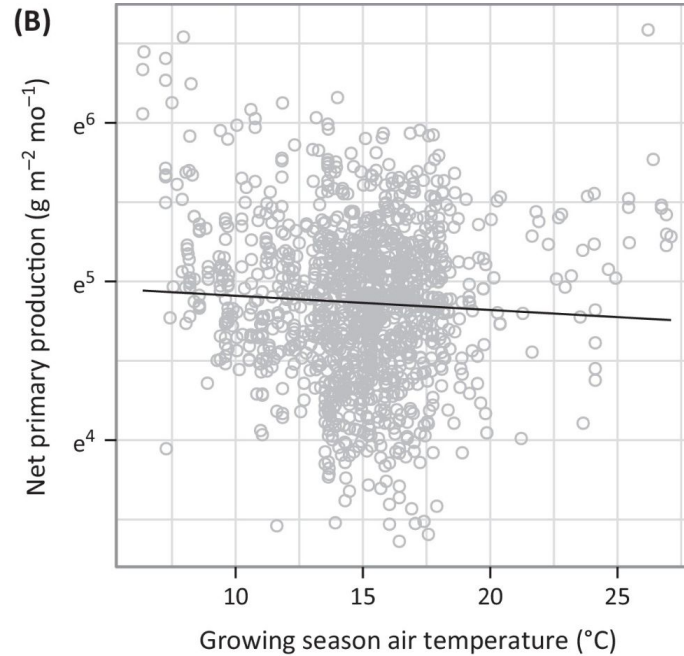
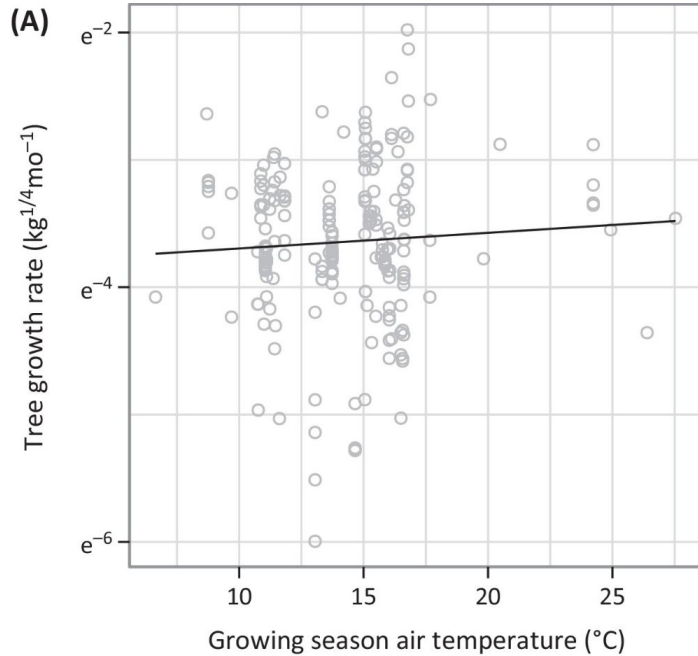
Bottom row is the percentage of the original $\Phi_{\text{excreta}} * \alpha B$ remaining. The error represents an uncertainty in extinct species distribution of 30%.

Opinion

Plant Thermoregulation:
Energetics, Trait–Environment
Interactions, and Carbon
Economics

Sean T. Michaletz,^{1,2,*} Michael D. Weiser,³ Jizhong Zhou,^{4,5,6}
Michael Kaspri,^{7,8} Brent R. Helliker,⁹ and
Brian J. Enquist^{1,9,10,11}

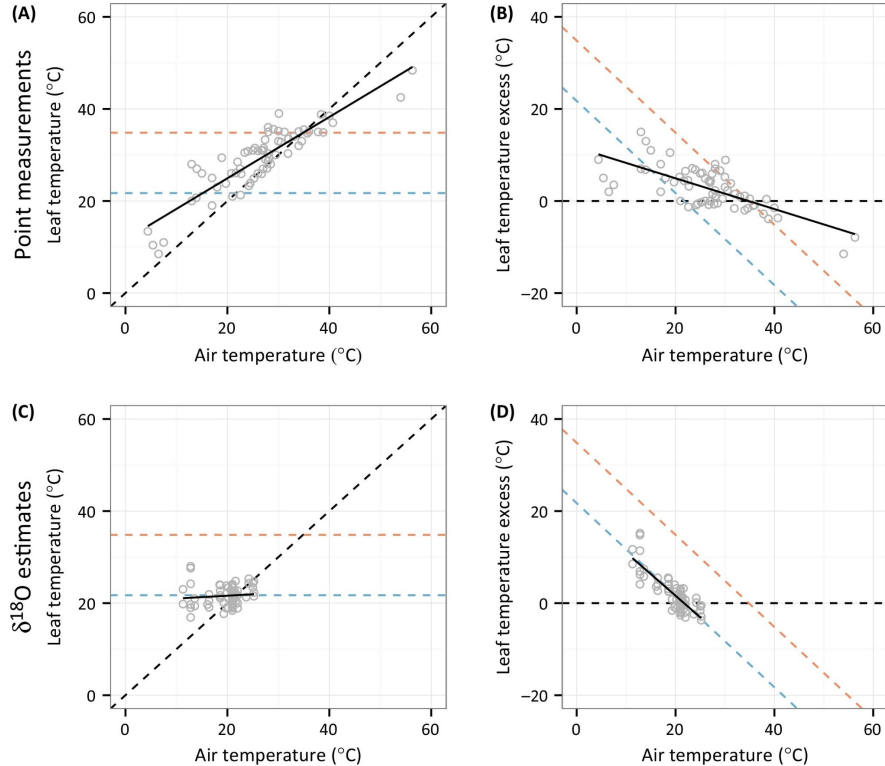
Traits and temp and photosynthesis



Trends in Ecology & Evolution

Photosynthesis is temp-dependent, but not much global signal of temperature

Traits and temp and photosynthesis



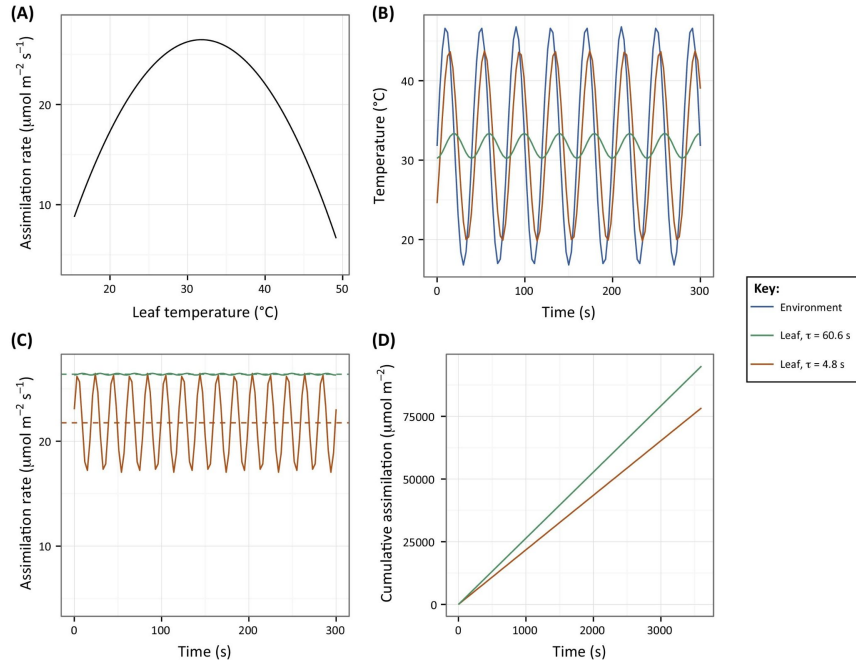
Opinion

Plant Thermoregulation: Energetics, Trait–Environment Interactions, and Carbon Economics

Sean T. Michaletz,^{1,2,*} Michael D. Weiser,³ Jizhong Zhou,^{4,5,6}
Michael Kaspari,^{7,8} Brent R. Helliker,⁹ and
Brian J. Enquist^{1,9,10,11}

Traits and temp and photosynthesis

If we assume two leaves sharing a common photosynthesis temperature response (A) and environment (B), but different thermal time constants τ (B)



Then energy budgets predict differences in time-averaged assimilation rates (C) and time-integrated carbon assimilation (D).

Opinion

Plant Thermoregulation: Energetics, Trait–Environment Interactions, and Carbon Economics

Sean T. Michaletz,^{1,2,*} Michael D. Weiser,³ Jizhong Zhou,^{4,5,6} Michael Kaspari,^{7,8} Brent R. Helliker,⁹ and Brian J. Enquist^{1,9,10,11}

Traits and temp and photosynthesis

We need to predict photosynthetic rate

Photosynthetic rate depends on temperature

Leaf temperature can't be predicted from air temp

How do we get at leaf temp?

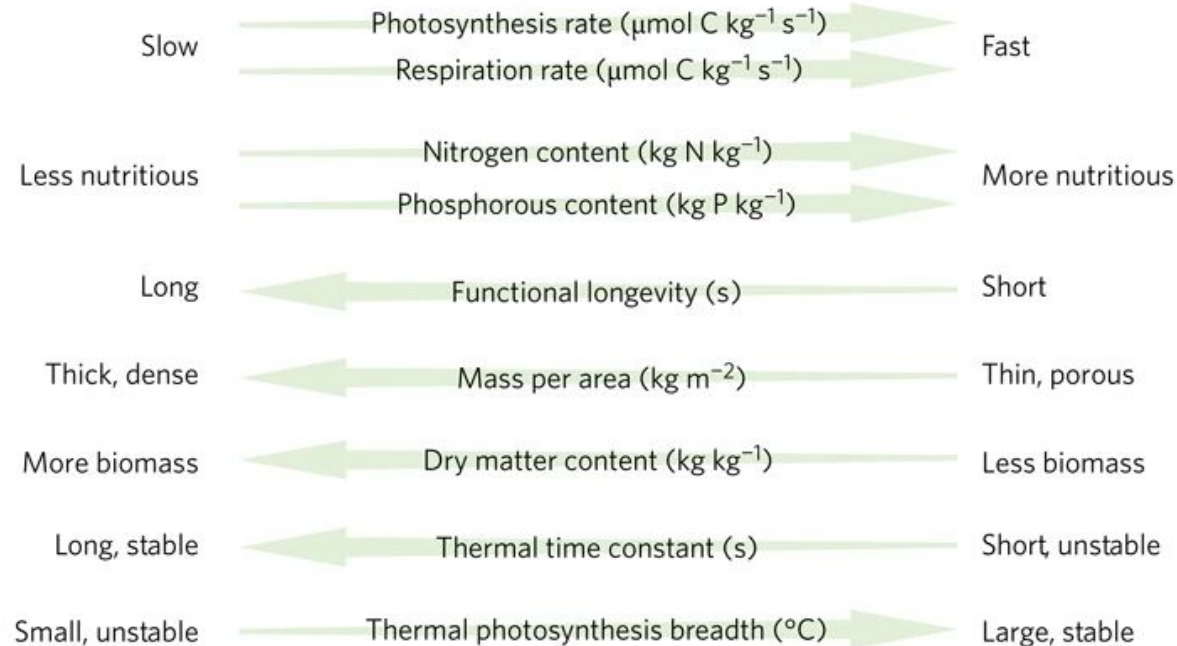
The energetic and carbon economic origins of leaf thermoregulation

Sean T. Michaletz^{1,2*}, Michael D. Weiser¹, Nate G. McDowell¹, Jizhong Zhou^{4,5,6}, Michael Kaspar^{3,7}, Brent R. Helliker⁸ and Brian J. Enquist^{2,9,10,11}

The energetic and carbon economic origins of leaf thermoregulation

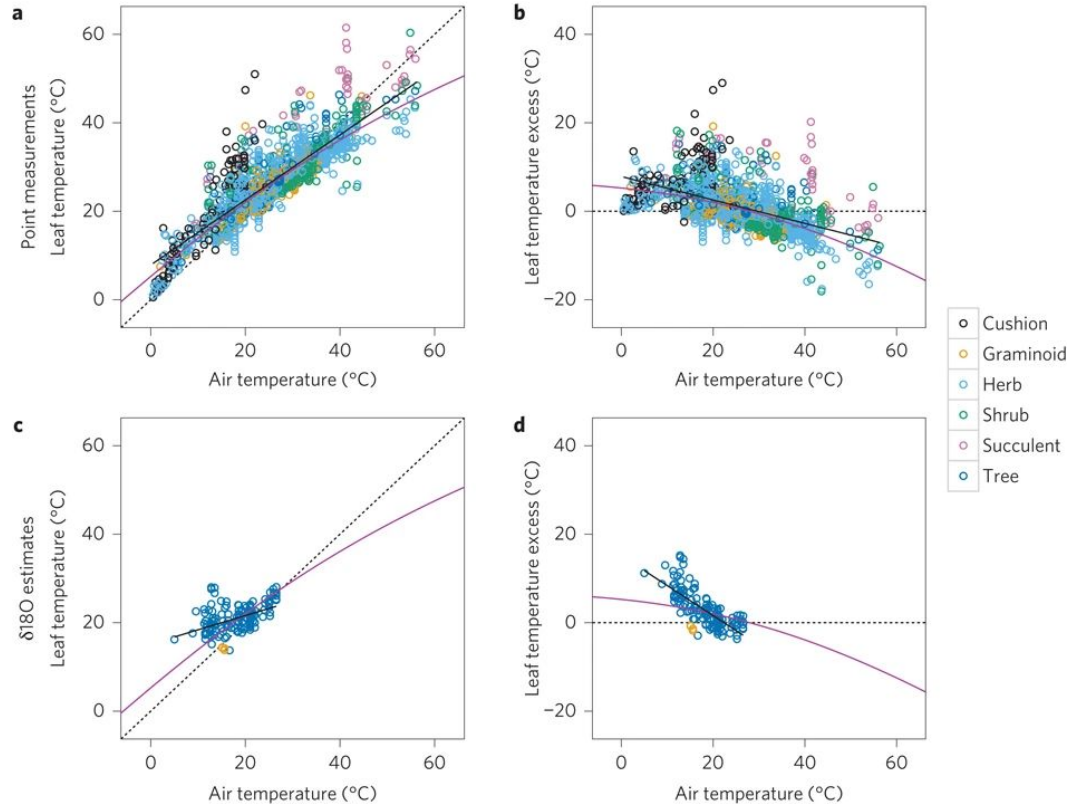
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Traits and temp and photosynthesis



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Pink = prediction from trait-based model

Black = best fit regression

Traits and temp and photosynthesis

Studies and theory like this represent major advances
in linking form and function

The energetic and carbon economic origins of leaf thermoregulation

Sean T. Michaletz^{1,2*}, Michael D. Weiser¹, Nate G. McDowell¹, Jizhong Zhou^{4,5,6}, Michael Kaspar^{1,7},
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Intraspecific variation matters

Many studies use species level mean trait values

- Turns out, this is a bad idea for estimating functioning

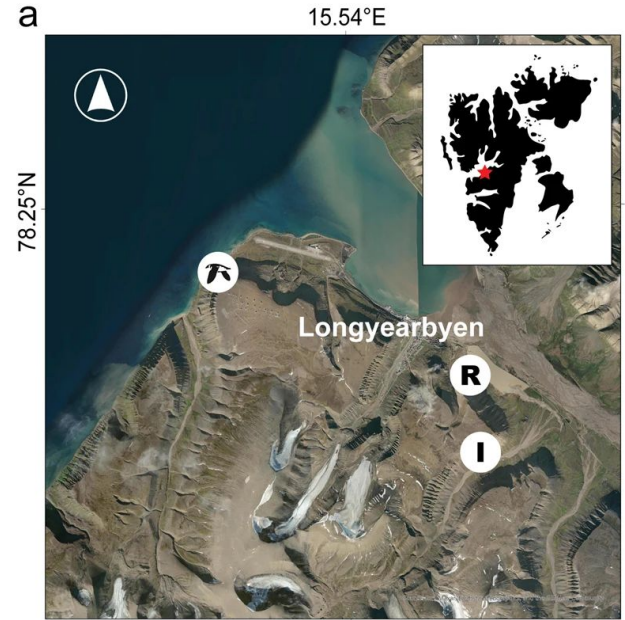
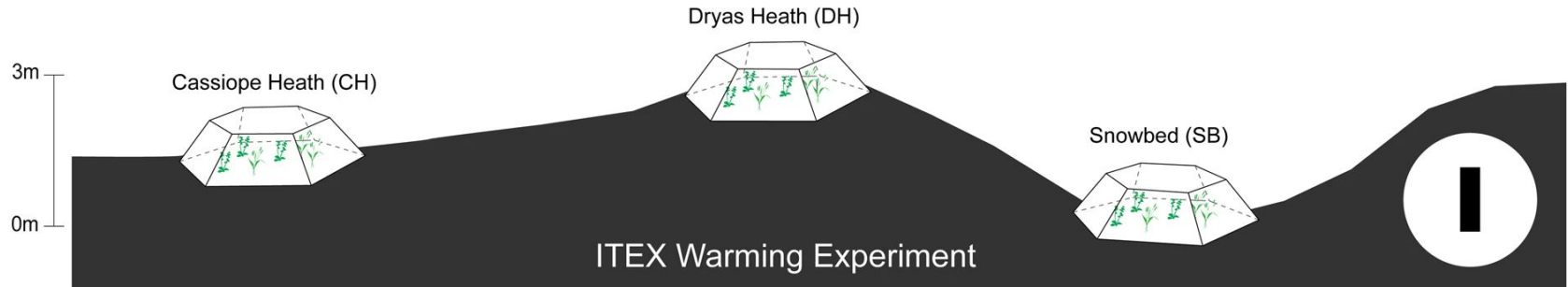
Intraspecific variation matters

Overview:

Measured CO₂ fluxes

Measured traits (including intraspecific variation)

Measured environment



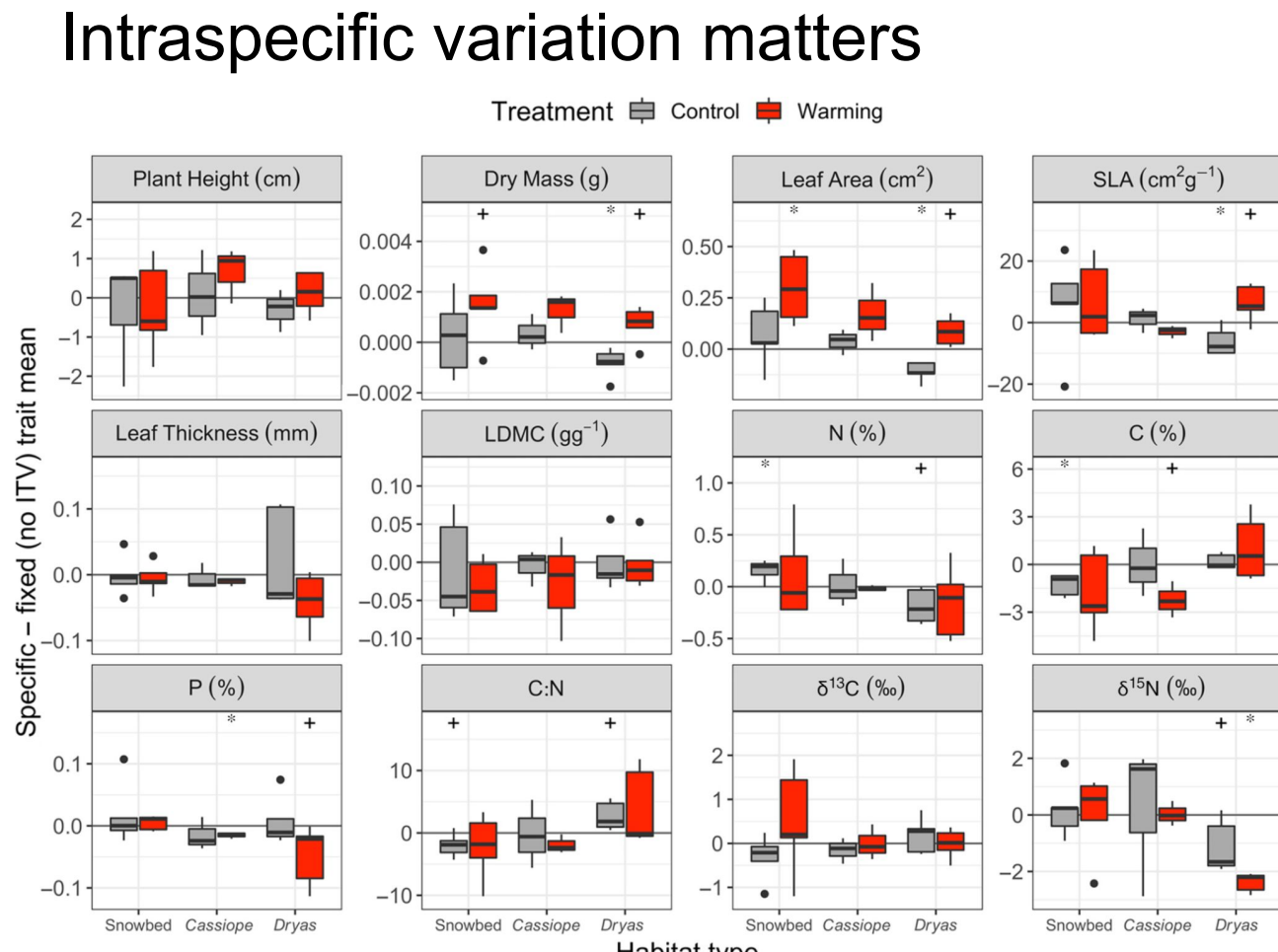
Intraspecific variation matters

To estimate the magnitude of ITV:

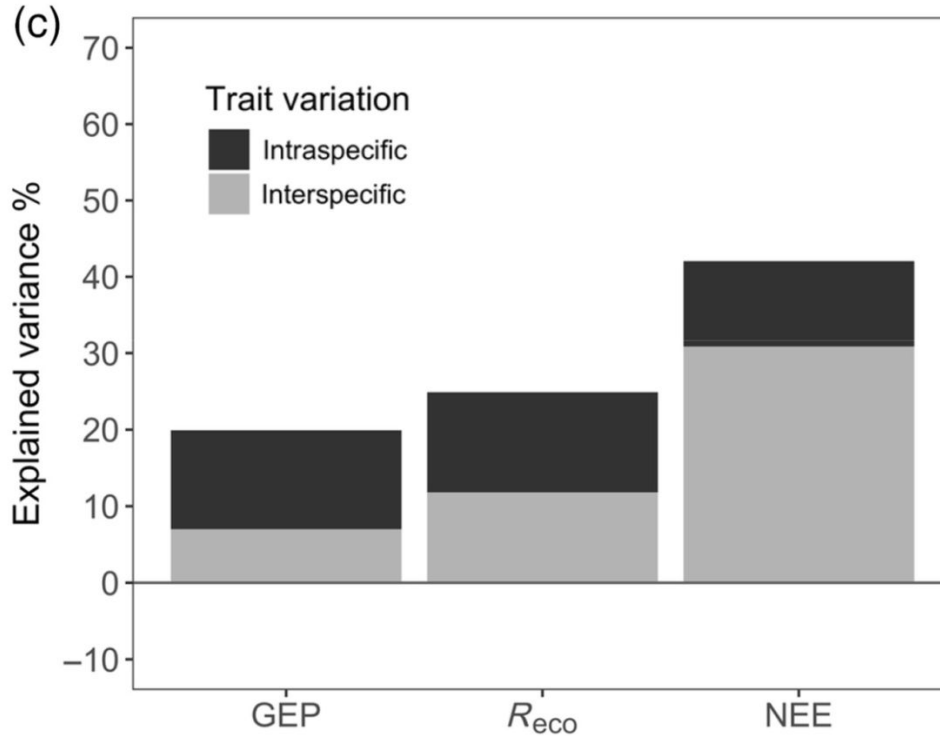
- Compare trait estimates using species mean vs those with ITV

Intraspecific trait variability is a key feature underlying high Arctic plant community resistance to climate warming

Ingibjörg S. Jónsdóttir^{1,2} | Aud H. Halbritter³ | Casper T. Christiansen^{4,5,6} | Inge H. J. Althuisen⁴ | Siri V. Haugum³ | Jonathan J. Henn^{7,8} | Katrin Björnsdóttir⁹ | Brian Salvin Maitner¹⁰ | Yadvinder Malhi¹¹ | Sean T. Michaletz¹² | Ruben E. Roos¹³ | Kari Klenderud¹³ | Hanna Lee^{5,14} | Brian J. Enquist¹⁵ | Vigdis Vandvik^{3,4}



Intraspecific variation matters



GEP = gross ecosystem productivity

R_{eco} = ecosystem respiration

NEE = net ecosystem CO_2 exchange

$GEP = NEE + R_{eco}$

Takeaways

- Traits help us understand variation in ecosystem services
- Traits - environment interactions can be important
- Mechanistic trait models can provide important insights
- Intraspecific trait variation is super important
- Some species are more important than others

Next class: Traits and global change